SUP'COM
Higher School of Communication of Tunis

MY VOICE
BIOMETRIC AUTHENTICATION

Major: NCA

Automatic Text Summarization

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- Work proposed and realised within: MyVoice.Ai

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3. Data Pre-processing
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1. Introduction

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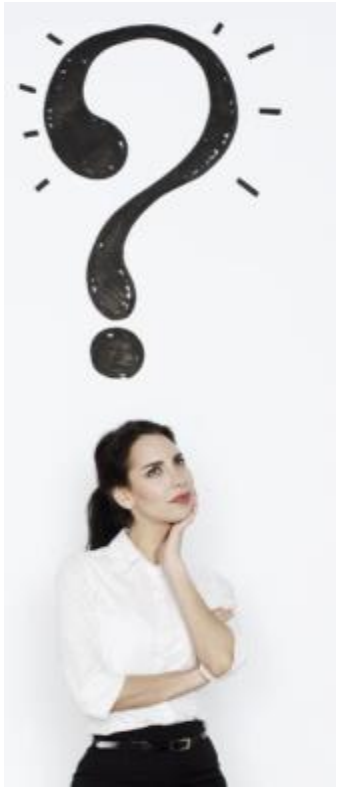
Results of Evaluation

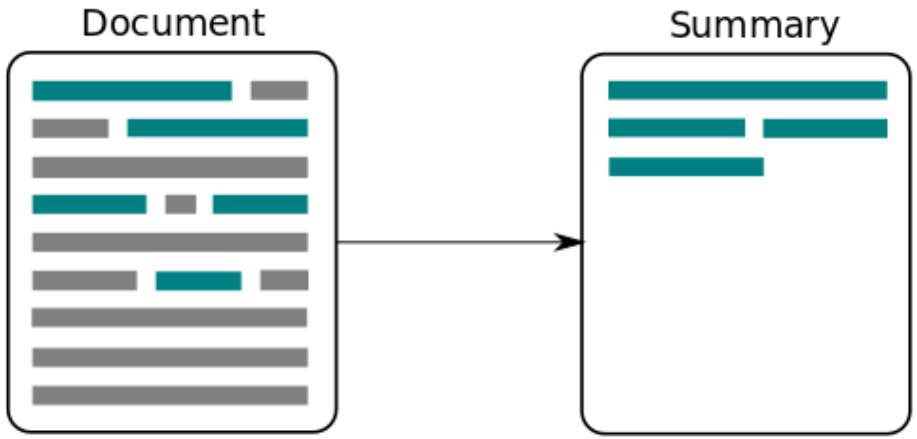
Conclusion and Perspectives

1-Introduction: What is Automatic Text Summarization?

Automatic text summarization:

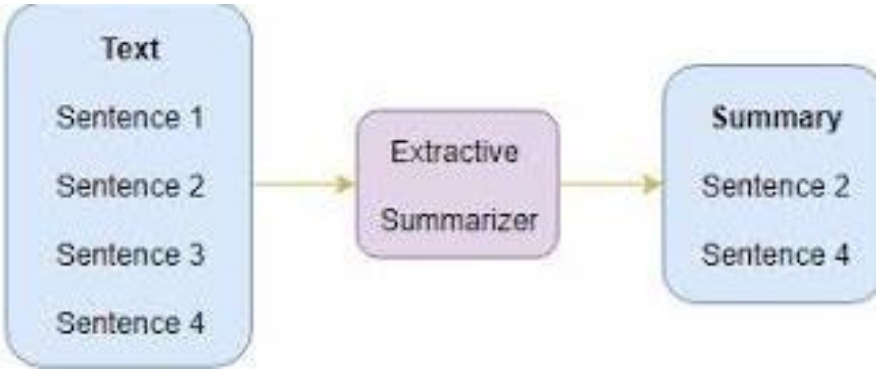
- ✓ Process of creating a short and coherent version of a longer document.
- ✓ Process of distilling the most important information from a source.





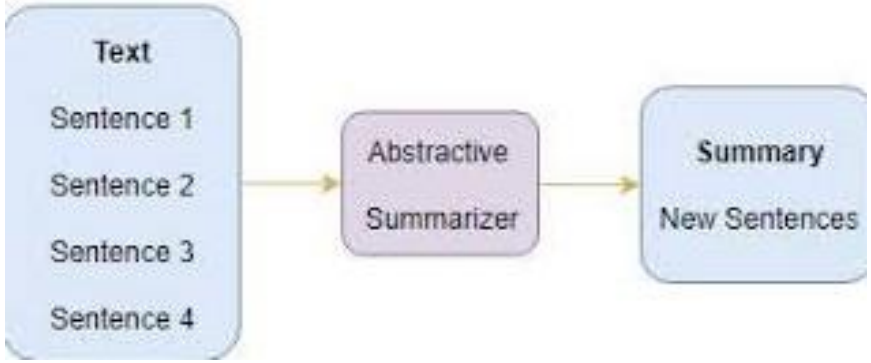
1-Introduction: How to Summarize a Document?

- Extractive methods: Extract sentences from the source document.



- Abstractive methods: Generate new sentences.





1-Introduction: Problematic Positioning

- ❖ Enormous amount of textual material:
  - ✓ Internet
  - ✓ Web pages
  - ✓ News articles





- ❖ Many summarization methods exist:
  - ✓ well-structured data
  - ✓ well-written data



We can not possibly create summaries of all of the text manually, automatic methods help us sort through large amount of data.

1-Introduction: Objectives

- Train a model to summarize text documents for extractive and abstractive tasks.
- Determine the most important one for summarization by calculating ROUGE scores.
- Apply the model that is trained to summarize text documents to summary transcribed speech documents.

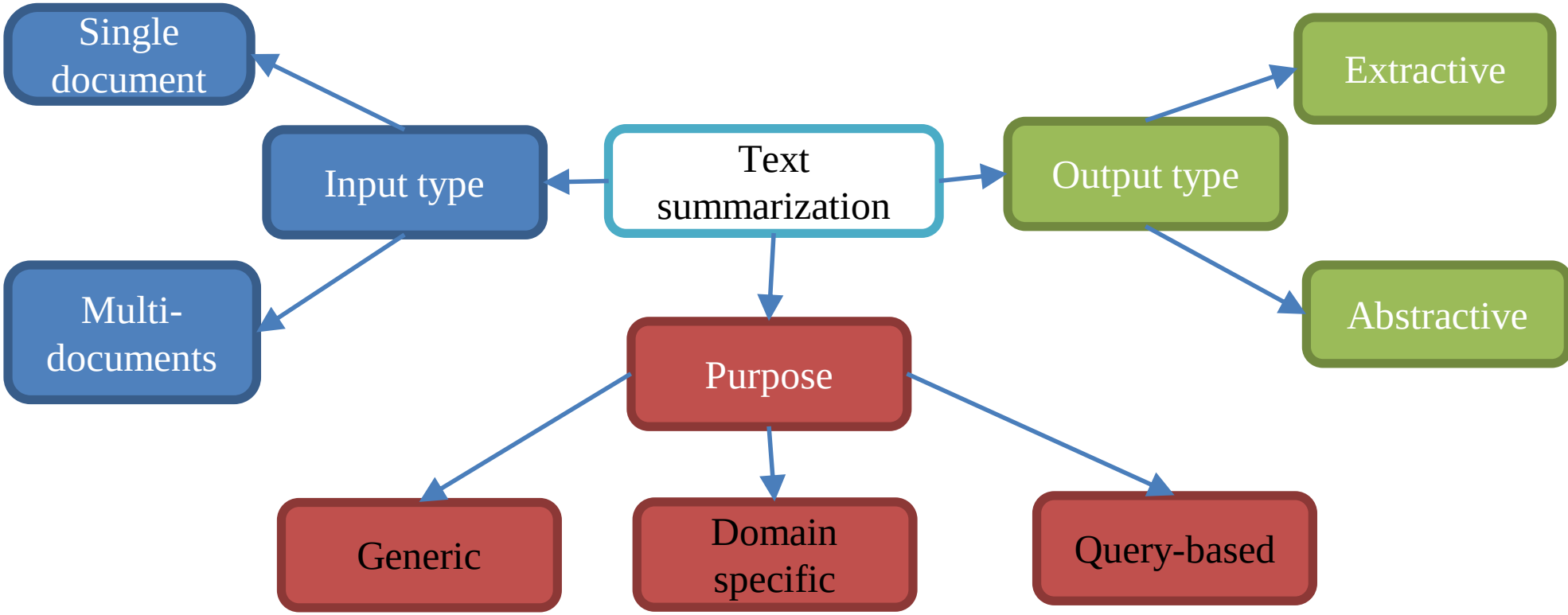


More data is currently available from text, and speech summarization is a less explored data.

1-Introduction: Summarization Factors:

Summarization Factors are:

- Input factors
- Purpose factors
- Output factors: Extractive or abstractive



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- ✓ ELMo (Embedding from Language Model): Semi-bidirectional
- ✓ GPT (generative pre-training) : Unidirectional

Why BERT?

- ✓ Purely Bi-directional
- ✓ ELMO is highly task dependent ➔ Depend on what it is used for.
- ✓ GPT is task-independent ➔ It is easy to fine-tune to new tasks.

BERT combines the best of ELMO and GPT:
➔ It is truly bidirectional and easy to fine-tune for new tasks.

2-Proposed Methods: Performance Evaluation

- ❑ Extrinsic
- ❑ Intrinsic ➔ ROUGE

$$F1 = 2 \cdot \frac{Recall \cdot Precision}{Recall + Precision}$$

Why ROUGE?

Extrinsic evaluation : - Costly.
- It takes human time.
- It requires some task that humans perform.

Intrinsic evaluations: - Faster.
- Less expensive.

- ROUGE-1
- ROUGE-2
- ROUGE-L

ROUGE compares the automatically generated summary and the reference summary.

2-Proposed Methods: Performance Evaluation

$$Recall = \frac{Number_of_overlapping_words}{Total_words_in_reference_summary}$$

$$Precision = \frac{Number_of_overlapping_words}{Total_words_in_system_summary}$$

- System Summary: The cat was found under the bed.
- Reference Summary: The cat was under the bed.

✓ For unigram: Recall= 6/6= 1
Precision=6/7= 0.86

✓ For bigram: Recall=4/5= 0.8
Precision=4/6= 0.67

✓ For LCS: Recall= 6/6=1. Precision=6/7=0.86

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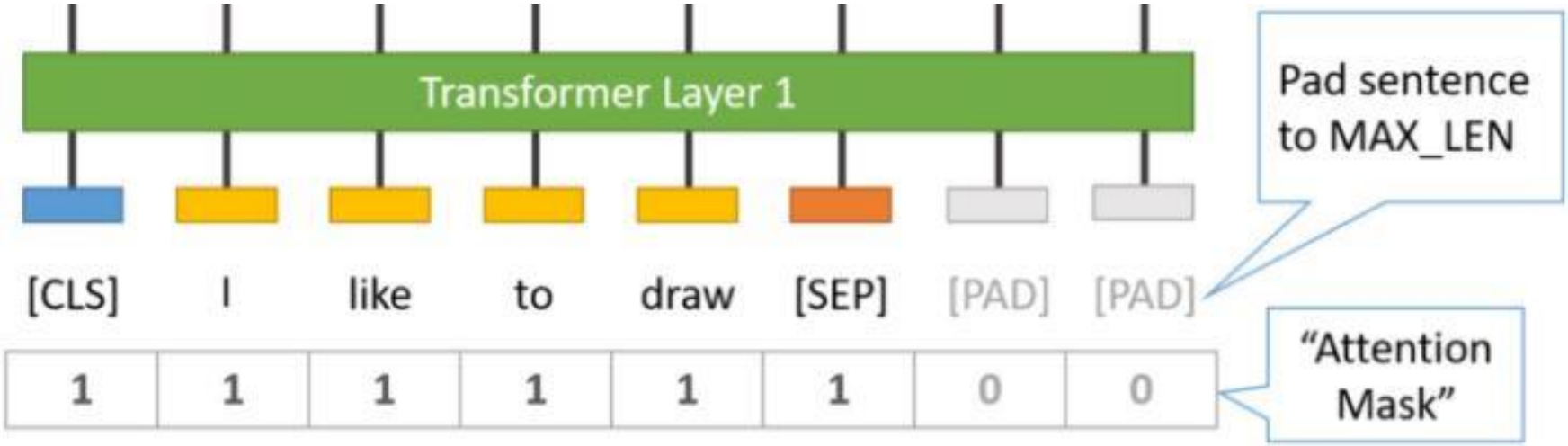
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3-Data Pre-processing: For Training

- ❖ CNN/DailyMail: 312085 documents ➔ Story format(text, highlights).
- ✓ BERT Tokenizer: Sentences ➔ Json format.
- ✓ Required Formatting: [CLS], [SEP], [PAD], attention mask.



The aim is to transform our dataset into the format that BERT can be trained on.

3-Data Pre-processing: For Testing and Evaluation

- Newsroom: Json lines format ➔ Plain text format ➔ **Extractive task**
- XSum(226711 news articles) ➔ **Abstractive task**
- AMI corpus(138 meetings): XML format ➔ TSV format (text+tabulation+summary)

➔ **Abstractive task**

The aim is to transform our dataset into the format that we can test and evaluate on.

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4- Training and Evaluation

❑ Training:

- ❑ Extractive and abstractive task.
- ❑ Extractive: 5000 steps with 2 GPU.
  - ✓ Model checkpoints saved every 1000 steps.
- ❑ Abstractive: 154000 steps with 2 GPU.
  - ✓ Model checkpoints saved every 2000 steps.

➔Abstractive takes more steps to train the model.

❑ Evaluation

✓ Extractive

```
[2020-05-31 21:48:01,283 INFO] Rouges at step 39000
>> ROUGE-F(1/2/3/l): 42.74/19.96/39.19
ROUGE-R(1/2/3/l): 53.12/24.78/48.66
```

✓ Abstractive

```
[2020-06-03 00:54:15,904 INFO] Rouges at step 120000
>> ROUGE-F(1/2/3/l): 39.78/17.94/36.98
ROUGE-R(1/2/3/l): 41.84/18.77/38.88
```

✓ Extractive ➔ Newsroom dataset and AMI corpus.

✓ Abstractive ➔ Xsum dataset.

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5-Results of Evaluation:

❑ Evaluation with CNN/DailMail

Model	R1	R2	RL
Extractive			
SUMMARUNNER	39.60	16.20	35.30
REFRESH	40.00	18.20	36.60
Abstractive			
PTGEN	36.44	15.66	33.42
DRM	39.87	15.82	36.90
Our model			
Extractive	**42.74**	**19.96**	**39.19**
Abstractive	39.78	17.94	36.98

Tab1: ROUGE F1 scores on CNN/DailyMail

5-Results of Evaluation:

❑ Evaluation with Newsroom

Model	R1	R2	RL
Extractive			
TextRank	22.76	9.8	18.97
Our model			
Extractive	**26.73**	**16.32**	**13.79**

Tab2: ROUGE F1 scores on Newsroom

5-Results of Evaluation:

❑ Evaluation with XSum

Model	R1	R2	RL
Abstractive			
PTGEN	29.7	9.21	23.24
Our model			
Abstractive	19.8	3.07	15.88

Tab3: ROUGE F1 scores on XSum

5-Results of Evaluation:

❑ Evaluation with AMI

Model	R1	R2	RL
Existing model(ami-ami)			
Abstractive	31.92	14.28	26.87
Our model(text-ami)			
Abstractive	22.91	6.46	19.87

Tab4: ROUGE F1 scores on AMI

5-Results of Evaluation:

❑ Analyses

- ❑ Calculate Perplexity .
- ✓ Smoothing technique ➔ Avoid zeros probabilities.
- ❑ Type of smoothing technique:
  - Add-one(Laplace smoothing)➔ Quick and easy.
  - Addalpha.
  - Add-delta(Lidstone smoothing),…

Understand similarities and differences in the data, for why the previous results (text BERT - speech data) seem to be mis-matched.

5-Results of Evaluation:

Train (AMI)

		Average(PPL)
Unigram	Test(AMI)	992.97
	Test(CNN/DailyMail)	6473.49
Bigram	Test(AMI)	322.76
	Test(CNN/DailyMail)	1185.35
Trigram	Test(AMI)	223.03
	Test(CNN/DailyMail)	627.28

- Lower PPL means better much.

Tab5: Results of average PPL

Average of perplexities of the model trained on AMI and tested on AMI or on CNN/DailyMail.

5-Results of Evaluation:

Train (CNN/DailyMail)

		Average(PPL)
Unigram	Test(AMI)	3594.44
	Test(CNN/DailyMail)	3699.37
Bigram	Test(AMI)	1228.73
	Test(CNN/DailyMail)	1596.53
Trigram	Test(AMI)	790.95
	Test(CNN/DailyMail)	1090.41

- The perplexity decrease as the n-gram size gets larger.

Tab6: Results of average PPL

Average of perplexities of the model trained on CNN/DailyMail and tested on AMI or on CNN/DailyMail

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✓ BERT can be usefully applied in text summarization for both extractive and abstractive task.

✓ BERT equally does not generalize well to other text datasets (Newsroom, Xsum), and transcribed speech data (AMI).

➔ BERT over-fits to the CNN dataset.



6-Conclusion and Perspectives: Perspectives

- ✓ BERT *base* model ➔ BERT *large* model .
- ✓ Adjust the BERT model ➔ Generalize better to other datasets.
- ✓ Fine-tune training on each specific dataset .
- ✓ Selectively choose subsets of sentences for training or testing that help the model become more robust.
- ✓ Use curriculum training or scheduled learning providing particular sentences to the training method at particular times during training.

➔We plain to expend the curriculum learning experiments and publish in the ACL conference.

Thank you!

Questions

